



CS102.3 - Programming with C

Group Assignment – Group No. 10

Bank Management System

Submission deadline: 20th February 2024

Ediriweera C.A.

Hearath H.M.T.J.

Jayasinghe K.U.

Weerasinghe E.M.L.C.

Weerasinghe M.G.H.

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1.Introduction:

The Bank Management System is an application for maintaining a person's account in a bank. The system provides the access to the costumer to create an account, deposit/withdraw the cash from his account, also to convert currency. The following documentation provides the specification of the system.

We are mainly concerned with developing a banking system where a Customer can submit his/her deposit amount to bank if he/she has an account or can create a new account in this bank. Customer can also view the status and change currency of his/her bank account, can view account balance. One can easily maintain the above things if he/she has an account by login through his unique account number

What we are built:-

In this project, we will build a small banking management system in C language.In which we can add accounts,search transactions and many more.



2.Project requirments:

2.1: ER Diagram :-

*Creating a detailed Entity-Relationship (ER) diagram for a bank management system involves identifying and representing various entities, their attributes, and the relationships between them. Below is a simplified ER diagram for a basic bank management system:

Entities:- 1.account

2.Transaction

3.Account holder

Attributes:- 1.Acc_balance

2.Acc_number

3.Holders_name

4.Transaction ID

5.Deposite amount

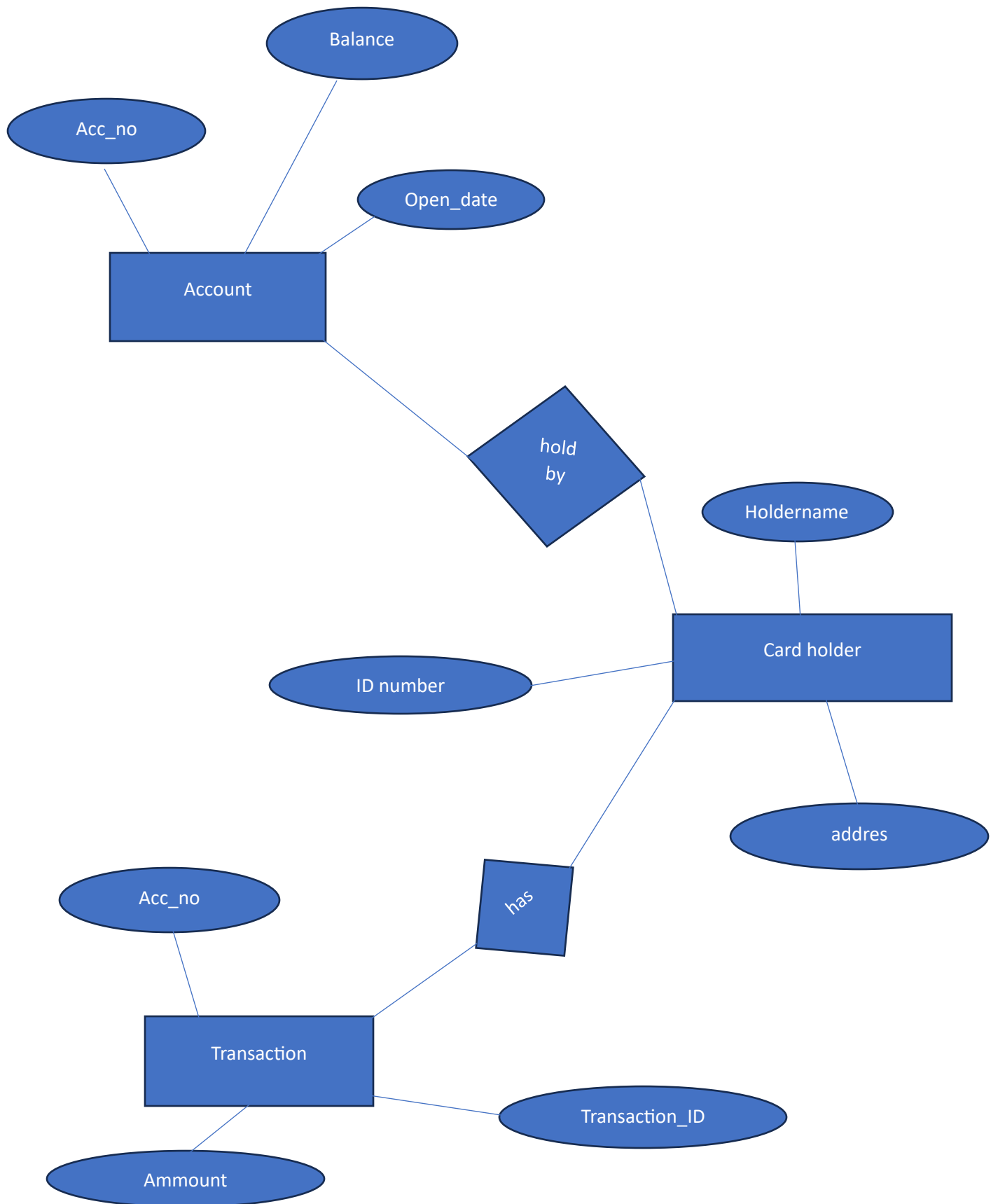
6.Withdrawal amount

Relations:- 1.hold by

2.has



Structure Of ER Diagram



2.2.Data Base

Pre requiments:-

- Header files
- Increment operator
- Functions
- Switch cases
- Basic understanding pf arrays

Header files:- Declare the function prototypes for functions defined in other source file

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

Increment operator:- basically used to keep track of count (**i ++**)

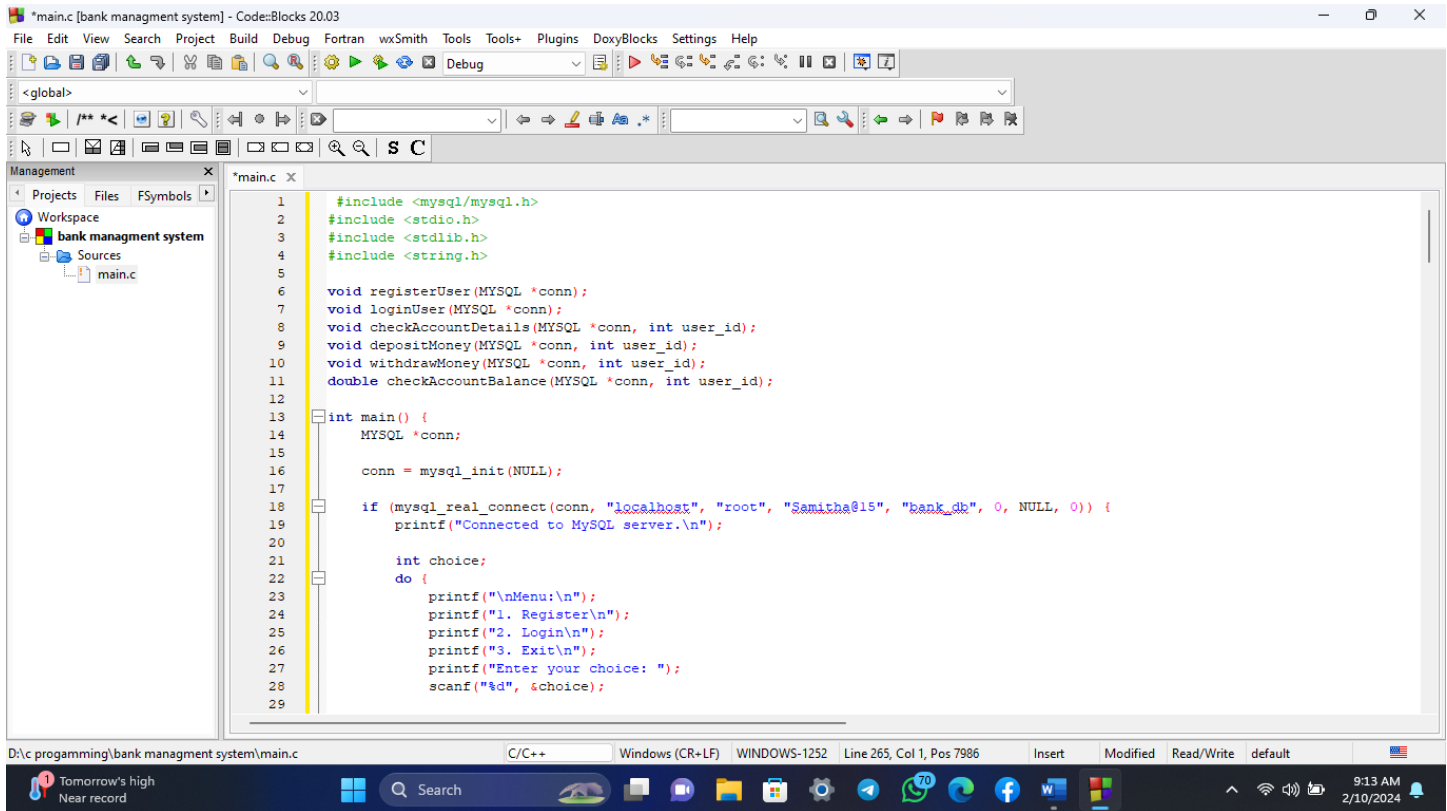
Function:- A block of code which only runs when it is called

Switch cases:-evaluates a given expression and based on the evaluated value, it executes the statements associated with it.

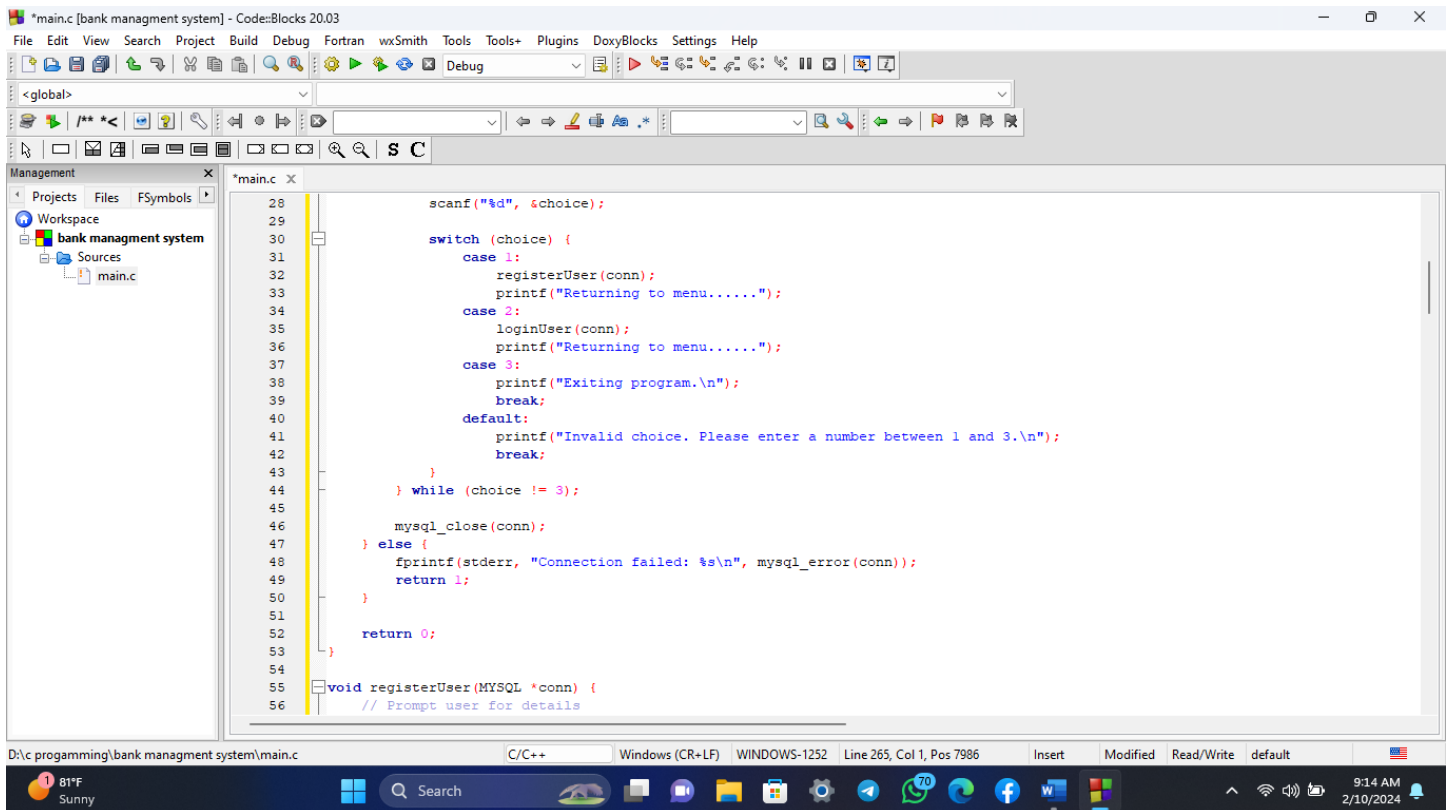
How are we going to build the database system:-

Our strategy will be to develop distinct functions for each activity which will then be combined to form the project's functional unit.

2.3.Screenshots Of Code:



```
*main.c [bank management system] - Code::Blocks 20.03
File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help
<global>
Management
  Projects Files FSymbols
  Workspace
  bank management system
    Sources
      main.c
*main.c X
1  #include <mysql/mysql.h>
2  #include <stdio.h>
3  #include <stdlib.h>
4  #include <string.h>
5
6  void registerUser(MYSQL *conn);
7  void loginUser(MYSQL *conn);
8  void checkAccountDetails(MYSQL *conn, int user_id);
9  void depositMoney(MYSQL *conn, int user_id);
10 void withdrawMoney(MYSQL *conn, int user_id);
11 double checkAccountBalance(MYSQL *conn, int user_id);
12
13 int main() {
14     MYSQL *conn;
15
16     conn = mysql_init(NULL);
17
18     if (mysql_real_connect(conn, "localhost", "root", "Samitha@15", "bank_db", 0, NULL, 0)) {
19         printf("Connected to MySQL server.\n");
20
21         int choice;
22         do {
23             printf("\nMenu:\n");
24             printf("1. Register\n");
25             printf("2. Login\n");
26             printf("3. Exit\n");
27             printf("Enter your choice: ");
28             scanf("%d", &choice);
29 }
```



```
*main.c [bank management system] - Code::Blocks 20.03
File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help
<global>
Management
  Projects Files FSymbols
  Workspace
  bank management system
    Sources
      main.c
*main.c X
28     scanf("%d", &choice);
29
30     switch (choice) {
31         case 1:
32             registerUser(conn);
33             printf("Returning to menu.....");
34         case 2:
35             loginUser(conn);
36             printf("Returning to menu.....");
37         case 3:
38             printf("Exiting program.\n");
39             break;
40         default:
41             printf("Invalid choice. Please enter a number between 1 and 3.\n");
42             break;
43     } while (choice != 3);
44
45     mysql_close(conn);
46 } else {
47     fprintf(stderr, "Connection failed: %s\n", mysql_error(conn));
48     return 1;
49 }
50
51 return 0;
52
53 void registerUser(MYSQL *conn) {
54     // Prompt user for details
55 }
```

*main.c [bank management system] - Code::Blocks 20.03

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<global>

Management

Projects Files FSymbols

Workspace

bank management system

Sources

main.c

```

52     return 0;
53 }
54
55 void registerUser(MYSQL *conn) {
56     // Prompt user for details
57     char name[256];
58     char address[256];
59     char phone_number[16];
60     char nic_number[16];
61     int pin;
62
63     printf("Enter Name: ");
64     scanf("%s", name);
65     printf("Enter Address: ");
66     scanf("%s", address);
67     printf("Enter Phone Number: ");
68     scanf("%s", phone_number);
69     printf("Enter NIC Number: ");
70     scanf("%s", nic_number);
71     printf("Enter PIN: ");
72     scanf("%d", &pin);
73
74     // Execute query to insert user details into your_table_name
75     char query[256];
76     sprintf(query, "INSERT INTO users (name, address, phone_number, nic_number) VALUES ('%s', '%s', '%s', '%s')",
77             name, address, phone_number, nic_number);
78
79     if (mysql_query(conn, query)) {
80         fprintf(stderr, "Insertion into users failed: %s\n", mysql_error(conn));

```

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*main.c [bank management system] - Code::Blocks 20.03

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<global>

Management

Projects Files FSymbols

Workspace

bank management system

Sources

main.c

```

70     scanf("%s", nic_number);
71     printf("Enter PIN: ");
72     scanf("%d", &pin);
73
74     // Execute query to insert user details into your_table_name
75     char query[256];
76     sprintf(query, "INSERT INTO users (name, address, phone_number, nic_number) VALUES ('%s', '%s', '%s', '%s')",
77             name, address, phone_number, nic_number);
78
79     if (mysql_query(conn, query)) {
80         fprintf(stderr, "Insertion into users failed: %s\n", mysql_error(conn));
81         return;
82     }
83
84     // Get the user_id of the newly inserted row
85     int user_id = mysql_insert_id(conn);
86
87     // Execute query to insert account details into account table
88     sprintf(query, "INSERT INTO account (user_id, balance, pin) VALUES (%d, 0.0, %d)", user_id, pin);
89
90     if (mysql_query(conn, query)) {
91         fprintf(stderr, "Insertion into account failed: %s\n", mysql_error(conn));
92         return;
93     }
94
95     printf("User registered successfully.\n");
96
97     printf("\nUser Details:\n");
98     printf("User ID: %d, Name: %s, Address: %s, Phone Number: %s, NIC Number: %s\n", user_id, name, address, phone_number, nic_nu

```

D:\c programming\bank management system\main.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 80, Col 81, Pos 2414 Insert Modified Read/Write default 81°F Sunny 9:14 AM 2/10/2024

*main.c [bank managment system] - Code::Blocks 20.03

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<global> registerUser(MYSQL* conn): void

```

91         fprintf(stderr, "Insertion into account failed: %s\n", mysql_error(conn));
92         return;
93     }
94
95     printf("User registered successfully.\n");
96
97     printf("\nUser Details:\n");
98     printf("User ID: %d, Name: %s, Address: %s, Phone Number: %s, NIC Number: %s\n", user_id, name, address, phone_number, nic_number);
99
100    printf("\nAccount Details:\n");
101    printf("Account ID: %d, User ID: %d, Balance: %.2f, PIN: %d\n", mysql_insert_id(conn), user_id, 0.0, pin);
102    printf("\nPlease Note down thease details.....!!!\n");
103 }
104
105 void loginUser(MYSQL *conn) {
106     // Prompt user for user_id and pin
107     int user_id, pin;
108
109     printf("Enter User ID: ");
110     scanf("%d", &user_id);
111     printf("Enter PIN: ");
112     scanf("%d", &pin);
113
114     // Execute query to check if user_id and pin match
115     char query[256];
116     sprintf(query, "SELECT * FROM account WHERE user_id = %d AND pin = %d", user_id, pin);
117
118     if (mysql_query(conn, query)) {
119         fprintf(stderr, "Login query failed: %s\n", mysql_error(conn));

```

D:\c programming\bank managment system\main.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 80, Col 81, Pos 2414 Insert Modified Read/Write default

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*main.c [bank managment system] - Code::Blocks 20.03

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<global> registerUser(MYSQL* conn): void

```

112     scanf("%d", &pin);
113
114     // Execute query to check if user_id and pin match
115     char query[256];
116     sprintf(query, "SELECT * FROM account WHERE user_id = %d AND pin = %d", user_id, pin);
117
118     if (mysql_query(conn, query)) {
119         fprintf(stderr, "Login query failed: %s\n", mysql_error(conn));
120         return;
121     }
122
123     MYSQL_RES *result = mysql_store_result(conn);
124     int num_rows = mysql_num_rows(result);
125
126     if (num_rows == 1) {
127         printf("Login successful.\n");
128
129         int choice;
130         do {
131             // Display menu for logged-in users
132             printf("\nLogged-In Menu:\n");
133             printf("1. Check Account Details\n");
134             printf("2. Deposit Money\n");
135             printf("3. Withdraw Money\n");
136             printf("4. Exit\n");
137             printf("Enter your choice: ");
138             scanf("%d", &choice);
139
140             // Process user choice

```

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*main.c [bank management system] - Code::Blocks 20.03

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<global> registerUser(MYSQL* conn): void

Management

Projects Files FSymbols

Workspace

bank management system

Sources

main.c

```

130         do {
131             // Display menu for logged-in users
132             printf("\nLogged-In Menu:\n");
133             printf("1. Check Account Details\n");
134             printf("2. Deposit Money\n");
135             printf("3. Withdraw Money\n");
136             printf("4. Exit\n");
137             printf("Enter your choice: ");
138             scanf("%d", &choice);
139
140             // Process user choice
141             switch (choice) {
142                 case 1:
143                     checkAccountDetails(conn, user_id);
144                     break;
145                 case 2:
146                     depositMoney(conn, user_id);
147                     break;
148                 case 3:
149                     withdrawMoney(conn, user_id);
150                     break;
151                 case 4:
152                     printf("Exiting logged-in menu.\n");
153                     break;
154                 default:
155                     printf("Invalid choice. Please enter a number between 1 and 4.\n");
156                     break;
157             }
158             while (choice != 4);

```

D:\c programming\bank management system\main.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 80, Col 81, Pos 2414 Insert Modified Read/Write default

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*main.c [bank management system] - Code::Blocks 20.03

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<global> registerUser(MYSQL* conn): void

Management

Projects Files FSymbols

Workspace

bank management system

Sources

main.c

```

145         case 2:
146             depositMoney(conn, user_id);
147             break;
148         case 3:
149             withdrawMoney(conn, user_id);
150             break;
151         case 4:
152             printf("Exiting logged-in menu.\n");
153             break;
154         default:
155             printf("Invalid choice. Please enter a number between 1 and 4.\n");
156             break;
157     }
158     while (choice != 4);
159 } else {
160     printf("Login failed. Invalid User ID or PIN.\n");
161 }
162 mysql_free_result(result);
163 }
164
165
166 void checkAccountDetails(MYSQL *conn, int user_id) {
167     char query[256];
168     sprintf(query, "SELECT * FROM account WHERE user_id = %d", user_id);
169
170     if (mysql_query(conn, query)) {
171         fprintf(stderr, "Check Account Details query failed: %s\n", mysql_error(conn));
172         return;
173     }

```

D:\c programming\bank management system\main.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 80, Col 81, Pos 2414 Insert Modified Read/Write default

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The screenshot shows a code editor window titled "main.c [bank management system] - Code::Blocks 20.03". The editor displays the implementation of the `checkAccountDetails` function. The function takes a MySQL connection pointer and a user ID as arguments. It constructs a SQL query to select account details for the given user ID. If the query fails, it prints an error message and returns. If successful, it fetches the result and prints the account details (ID, User ID, Balance, PIN) if found, or a message stating that details were not found. The function also calls `mysql_free_result` to free the result set.

```
166 void checkAccountDetails(MYSQL *conn, int user_id) {
167     char query[256];
168     sprintf(query, "SELECT * FROM account WHERE user_id = %d", user_id);
169
170     if (mysql_query(conn, query)) {
171         fprintf(stderr, "Check Account Details query failed: %s\n", mysql_error(conn));
172         return;
173     }
174
175     MYSQL_RES *result = mysql_store_result(conn);
176     MYSQL_ROW row = mysql_fetch_row(result);
177
178     if (row) {
179         printf("\nAccount Details:\n");
180         printf("Account ID: %s, User ID: %s, Balance: %s, PIN: %s\n", row[0], row[1], row[2], row[3]);
181     } else {
182         printf("Account details not found for the user.\n");
183     }
184
185     mysql_free_result(result);
186 }
187
188 void depositMoney(MYSQL *conn, int user_id) {
189     double amount;
190
191     printf("Enter the amount to deposit: ");
192     scanf("%lf", &amount);
193
194 }
```

The screenshot shows a code editor window titled "main.c [bank management system] - Code::Blocks 20.03". The editor displays the implementation of the `depositMoney` and `withdrawMoney` functions. The `depositMoney` function prompts the user for an amount to deposit. If the amount is less than or equal to zero, it prints an error message and returns. If the amount is positive, it constructs a SQL query to update the account balance by adding the deposited amount. If the query fails, it prints an error message and returns. If successful, it prints a message indicating the successful deposit and the updated balance. The `withdrawMoney` function prompts the user for an amount to withdraw.

```
187 }
188
189 void depositMoney(MYSQL *conn, int user_id) {
190     double amount;
191
192     printf("Enter the amount to deposit: ");
193     scanf("%lf", &amount);
194
195     if (amount <= 0) {
196         printf("Invalid deposit amount.\n");
197         return;
198     }
199
200     // Update the account balance
201     char query[256];
202     sprintf(query, "UPDATE account SET balance = balance + %.2lf WHERE user_id = %d", amount, user_id);
203
204     if (mysql_query(conn, query)) {
205         fprintf(stderr, "Deposit Money query failed: %s\n", mysql_error(conn));
206         return;
207     }
208
209     printf("Deposit successful. Updated balance: %.2lf\n", checkAccountBalance(conn, user_id));
210 }
211
212 void withdrawMoney(MYSQL *conn, int user_id) {
213     double amount;
214
215     printf("Enter the amount to withdraw: ");
216 }
```

*main.c [bank managment system] - Code::Blocks 20.03

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<global> registerUser(MYSQL* conn): void

```

211 void withdrawMoney(MYSQL *conn, int user_id) {
212     double amount;
213
214     printf("Enter the amount to withdraw: ");
215     scanf("%lf", &amount);
216
217     if (amount <= 0) {
218         printf("Invalid withdrawal amount.\n");
219         return;
220     }
221
222     // Check if there is sufficient balance
223     double currentBalance = checkAccountBalance(conn, user_id);
224
225     if (amount > currentBalance) {
226         printf("Insufficient funds. Cannot withdraw more than the current balance.\n");
227         return;
228     }
229
230     // Update the account balance
231     char query[256];
232     sprintf(query, "UPDATE account SET balance = balance - %.2lf WHERE user_id = %d", amount, user_id);
233
234     if (mysql_query(conn, query)) {
235         fprintf(stderr, "Withdraw Money query failed: %s\n", mysql_error(conn));
236         return;
237     }
238
239

```

D:\c programming\bank managment system\main.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 80, Col 81, Pos 2414 Insert Modified Read/Write default

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*main.c [bank managment system] - Code::Blocks 20.03

File Edit View Search Project Build Debug Fortran wxSmith Tools Tools+ Plugins DoxyBlocks Settings Help

<global> registerUser(MYSQL* conn): void

```

229
230
231 // Update the account balance
232 char query[256];
233 sprintf(query, "UPDATE account SET balance = balance - %.2lf WHERE user_id = %d", amount, user_id);
234
235 if (mysql_query(conn, query)) {
236     fprintf(stderr, "Withdraw Money query failed: %s\n", mysql_error(conn));
237     return;
238 }
239
240 printf("Withdrawal successful. Updated balance: %.2lf\n", checkAccountBalance(conn, user_id));
241
242
243 double checkAccountBalance(MYSQL *conn, int user_id) {
244     char query[256];
245     sprintf(query, "SELECT balance FROM account WHERE user_id = %d", user_id);
246
247     if (mysql_query(conn, query)) {
248         fprintf(stderr, "Check Account Balance query failed: %s\n", mysql_error(conn));
249         return -1.0;
250     }
251
252     MYSQL_RES *result = mysql_store_result(conn);
253     MYSQL_ROW row = mysql_fetch_row(result);
254
255     double balance = -1.0;
256
257     if (row) {

```

D:\c programming\bank managment system\main.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 80, Col 81, Pos 2414 Insert Modified Read/Write default

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*main.c [bank management system] - Code::Blocks 20.03

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<global> registerUser(MYSQL *conn): void

Management

Projects Files FSymbols

Workspace

bank management system

Sources

main.c

```
238 }
239
240 printf("Withdrawal successful. Updated balance: %.2f\n", checkAccountBalance(conn, user_id));
241
242
243 double checkAccountBalance(MYSQL *conn, int user_id) {
244     char query[256];
245     sprintf(query, "SELECT balance FROM account WHERE user_id = %d", user_id);
246
247     if (mysql_query(conn, query)) {
248         fprintf(stderr, "Check Account Balance query failed: %s\n", mysql_error(conn));
249         return -1.0;
250     }
251
252     MYSQL_RES *result = mysql_store_result(conn);
253     MYSQL_ROW row = mysql_fetch_row(result);
254
255     double balance = -1.0;
256
257     if (row) {
258         balance = atof(row[0]);
259     }
260
261     mysql_free_result(result);
262
263     return balance;
264 }
265
266
```

D:\c programming\bank management system\main.c C/C++ Windows (CR+LF) WINDOWS-1252 Line 80, Col 81, Pos 2414 Insert Modified Read/Write default

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2.4.Screenshots of Output:

```
tirani@tirani:~/Documents/BankingSystem$ ./main
Connected to MySQL server.
```

```
Menu:
1. Register
2. Login
3. Exit
Enter your choice: █
```

```
Menu:
1. Register
2. Login
3. Exit
Enter your choice: 1
Enter Name: tirani
Enter Address: homagama
Enter Phone Number: 0777555666
Enter NIC Number: 123456789
Enter PIN: 0000
User registered successfully.

User Details:
User ID: 4, Name: tirani, Address: homagama, Phone Number: 0777555666, NIC Number: 123456789

Account Details:
Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Please Notedown thease details.....!!!!
Returning to menu.....Enter User ID: █
```

```
tirani@tirani:~/Documents/BankingSystem$ ./main
Connected to MySQL server.

Menu:
1. Register
2. Login
3. Exit
Enter your choice: 1
Enter Name: tirani
Enter Address: homagama
Enter Phone Number: 0777555666
Enter NIC Number: 123456789
Enter PIN: 0000
User registered successfully.

User Details:
User ID: 4, Name: tirani, Address: homagama, Phone Number: 0777555666, NIC Number: 123456789

Account Details:
Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Please Notedown thease details.....!!!!
Returning to menu.....Enter User ID: 4
Enter PIN: 0000
Login successful.

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: █
```

```
tirani@tirani:~/Documents/BankingSystem$ ./main
Connected to MySQL server.
```

Menu:

1. Register
2. Login
3. Exit

Enter your choice: 1

Enter Name: tirani

Enter Address: homagana

Enter Phone Number: 0777555666

Enter NIC Number: 123456789

Enter PIN: 0000

User registered successfully.

User Details:

User ID: 4, Name: tirani, Address: homagana, Phone Number: 0777555666, NIC Number: 123456789

Account Details:

Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Please Notedown thease details.....!!!!

Returning to menu.....Enter User ID: 4

Enter PIN: 0000

Login successful.

Logged-In Menu:

1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit

Enter your choice: 1

Account Details:

Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Logged-In Menu:

1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit

Enter your choice: 2

Enter the amount to deposit: 25000

Deposit successful. Updated balance: 25000.00

Logged-In Menu:

1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit

Enter your choice: █

2. Login

3. Exit

Enter your choice: 1

Enter Name: tirani

Enter Address: homagana

Enter Phone Number: 0777555666

Enter NIC Number: 123456789

Enter PIN: 0000

User registered successfully.

User Details:

User ID: 4, Name: tirani, Address: homagana, Phone Number: 0777555666, NIC Number: 123456789

Account Details:

Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Please Notedown thease details.....!!!!

Returning to menu.....Enter User ID: 4

Enter PIN: 0000

Login successful.

Logged-In Menu:

1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit

Enter your choice: 1

Account Details:

Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Logged-In Menu:

1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit

Enter your choice: 2

Enter the amount to deposit: 25000

Deposit successful. Updated balance: 25000.00

Logged-In Menu:

1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit

Enter your choice: 3

Enter the amount to withdraw: 5000

Withdrawal successful. Updated balance: 20000.00

Logged-In Menu:

1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit

Enter your choice: █


```

Enter PIN: 0000
Login successful.

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 1

Account Details:
Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 2
Enter the amount to deposit: 25000
Deposit successful. Updated balance: 25000.00

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 3
Enter the amount to withdraw: 5000
Withdrawal successful. Updated balance: 20000.00

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 4
Exiting logged-in menu.
Returning to menu.....Exiting program.

Menu:
1. Register
2. Login
3. Exit
Enter your choice: 2
Enter User ID: 4
Enter PIN: 0000
Login successful.

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: █

```

```

Account Details:
Account ID: 10004, User ID: 4, Balance: 0.00, PIN: 0

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 2
Enter the amount to deposit: 25000
Deposit successful. Updated balance: 25000.00

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 3
Enter the amount to withdraw: 5000
Withdrawal successful. Updated balance: 20000.00

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 4
Exiting logged-in menu.
Returning to menu.....Exiting program.

Menu:
1. Register
2. Login
3. Exit
Enter your choice: 2
Enter User ID: 4
Enter PIN: 0000
Login successful.

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: 1

Account Details:
Account ID: 10004, User ID: 4, Balance: 20000.00, PIN: 0

Logged-In Menu:
1. Check Account Details
2. Deposit Money
3. Withdraw Money
4. Exit
Enter your choice: █

```



```
tirani@tirani:~/Documents/BankingSystem$ ./main
Connected to MySQL server.
```

```
Menu:
```

```
1. Register
```

```
2. Login
```

```
3. Exit
```

```
Enter your choice: █
```

3. Best Practices Utilized:

3.1. Code Structure:-

Designing a complete bank management system involves multiple components and functionalities. Below is a simplified example of a C program structure for a basic bank management system. This example includes structures for customer information, account details, and functions to perform common banking operations. Keep in mind that this is a basic example, and a real-world bank management system would be much more complex.

```
#include <mysql/mysql.h>

#include <stdio.h>

#include <stdlib.h>

#include <string.h>


void registerUser(MYSQL *conn);

void loginUser(MYSQL *conn);

void checkAccountDetails(MYSQL *conn, int user_id);

void depositMoney(MYSQL *conn, int user_id);

void withdrawMoney(MYSQL *conn, int user_id);

double checkAccountBalance(MYSQL *conn, int user_id);


int main() {

    MYSQL *conn;


    conn = mysql_init(NULL);


    if (mysql_real_connect(conn, "localhost", "root", "Samitha@15", "bank_db", 0, NULL, 0)) {

        printf("Connected to MySQL server.\n");


        int choice;

        do {

            printf("\nMenu:\n");

            printf("1. Register\n");

            printf("2. Login\n");
```

```

printf("3. Exit\n");

printf("Enter your choice: ");

scanf("%d", &choice);


switch (choice) {

    case 1:

        registerUser(conn);

        printf("Returning to menu.....");

    case 2:

        loginUser(conn);

        printf("Returning to menu.....");

    case 3:

        printf("Exiting program.\n");

        break;

    default:

        printf("Invalid choice. Please enter a number between 1 and 3.\n");

        break;

}

} while (choice != 3);


mysql_close(conn);

} else {

    fprintf(stderr, "Connection failed: %s\n", mysql_error(conn));

    return 1;

}


return 0;

}


void registerUser(MYSQL *conn) {

    // Prompt user for details

```

```

char name[256];
char address[256];
char phone_number[16];
char nic_number[16];
int pin;

printf("Enter Name: ");
scanf("%s", name);
printf("Enter Address: ");
scanf("%s", address);
printf("Enter Phone Number: ");
scanf("%s", phone_number);
printf("Enter NIC Number: ");
scanf("%s", nic_number);
printf("Enter PIN: ");
scanf("%d", &pin);

// Execute query to insert user details into your_table_name
char query[256];
sprintf(query, "INSERT INTO users (name, address, phone_number, nic_number) VALUES ('%s', '%s', '%s', '%s')",
        name, address, phone_number, nic_number);

if (mysql_query(conn, query)) {
    fprintf(stderr, "Insertion into users failed: %s\n", mysql_error(conn));
    return;
}

// Get the user_id of the newly inserted row
int user_id = mysql_insert_id(conn);

// Execute query to insert account details into account table

```

```

sprintf(query, "INSERT INTO account (user_id, balance, pin) VALUES (%d, 0.0, %d)", user_id, pin);

if (mysql_query(conn, query)) {
    fprintf(stderr, "Insertion into account failed: %s\n", mysql_error(conn));
    return;
}

printf("User registered successfully.\n");

printf("\nUser Details:\n");

printf("User ID: %d, Name: %s, Address: %s, Phone Number: %s, NIC Number: %s\n", user_id, name, address,
phone_number, nic_number);

printf("\nAccount Details:\n");

printf("Account ID: %d, User ID: %d, Balance: %.2f, PIN: %d\n", mysql_insert_id(conn), user_id, 0.0, pin);

printf("\nPlease Notedown thease details.....!!!\n");
}

void loginUser(MYSQL *conn) {
    // Prompt user for user_id and pin
    int user_id, pin;

    printf("Enter User ID: ");
    scanf("%d", &user_id);
    printf("Enter PIN: ");
    scanf("%d", &pin);

    // Execute query to check if user_id and pin match
    char query[256];
    sprintf(query, "SELECT * FROM account WHERE user_id = %d AND pin = %d", user_id, pin);

```

```
if (mysql_query(conn, query)) {  
    fprintf(stderr, "Login query failed: %s\n", mysql_error(conn));  
    return;  
}
```

```
MYSQL_RES *result = mysql_store_result(conn);  
int num_rows = mysql_num_rows(result);
```

```
if (num_rows == 1) {  
    printf("Login successful.\n");
```

```
int choice;
```

```
do {
```

```
    // Display menu for logged-in users
```

```
    printf("\nLogged-In Menu:\n");
```

```
    printf("1. Check Account Details\n");
```

```
    printf("2. Deposit Money\n");
```

```
    printf("3. Withdraw Money\n");
```

```
    printf("4. Exit\n");
```

```
    printf("Enter your choice: ");
```

```
    scanf("%d", &choice);
```

```
    // Process user choice
```

```
    switch (choice) {
```

```
        case 1:
```

```
            checkAccountDetails(conn, user_id);
```

```
            break;
```

```
        case 2:
```

```
            depositMoney(conn, user_id);
```

```
            break;
```

```
        case 3:
```

```

        withdrawMoney(conn, user_id);

        break;
    case 4:
        printf("Exiting logged-in menu.\n");
        break;
    default:
        printf("Invalid choice. Please enter a number between 1 and 4.\n");
        break;
    }
} while (choice != 4);
} else {
    printf("Login failed. Invalid User ID or PIN.\n");
}

mysql_free_result(result);
}

void checkAccountDetails(MYSQL *conn, int user_id) {
    char query[256];
    sprintf(query, "SELECT * FROM account WHERE user_id = %d", user_id);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Check Account Details query failed: %s\n", mysql_error(conn));
        return;
    }

    MYSQL_RES *result = mysql_store_result(conn);
    MYSQL_ROW row = mysql_fetch_row(result);

    if (row) {

```

```

    printf("\nAccount Details:\n");

    printf("Account ID: %s, User ID: %s, Balance: %s, PIN: %s\n", row[0], row[1], row[2], row[3]);
} else {
    printf("Account details not found for the user.\n");
}

mysql_free_result(result);
}

void depositMoney(MYSQL *conn, int user_id) {
    double amount;

    printf("Enter the amount to deposit: ");
    scanf("%lf", &amount);

    if (amount <= 0) {
        printf("Invalid deposit amount.\n");
        return;
    }

    // Update the account balance
    char query[256];
    sprintf(query, "UPDATE account SET balance = balance + %.2lf WHERE user_id = %d", amount, user_id);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Deposit Money query failed: %s\n", mysql_error(conn));
        return;
    }

    printf("Deposit successful. Updated balance: %.2lf\n", checkAccountBalance(conn, user_id));
}

```



```

void withdrawMoney(MYSQL *conn, int user_id) {
    double amount;

    printf("Enter the amount to withdraw: ");
    scanf("%lf", &amount);

    if (amount <= 0) {
        printf("Invalid withdrawal amount.\n");
        return;
    }

    // Check if there is sufficient balance
    double currentBalance = checkAccountBalance(conn, user_id);

    if (amount > currentBalance) {
        printf("Insufficient funds. Cannot withdraw more than the current balance.\n");
        return;
    }

    // Update the account balance
    char query[256];
    sprintf(query, "UPDATE account SET balance = balance - %.2lf WHERE user_id = %d", amount, user_id);

    if (mysql_query(conn, query)) {
        fprintf(stderr, "Withdraw Money query failed: %s\n", mysql_error(conn));
        return;
    }

    printf("Withdrawal successful. Updated balance: %.2lf\n", checkAccountBalance(conn, user_id));
}

```

```
double checkAccountBalance(MYSQL *conn, int user_id) {  
    char query[256];  
    sprintf(query, "SELECT balance FROM account WHERE user_id = %d", user_id);  
  
    if (mysql_query(conn, query)) {  
        fprintf(stderr, "Check Account Balance query failed: %s\n", mysql_error(conn));  
        return -1.0;  
    }  
  
    MYSQL_RES *result = mysql_store_result(conn);  
    MYSQL_ROW row = mysql_fetch_row(result);  
  
    double balance = -1.0;  
  
    if (row) {  
        balance = atof(row[0]);  
    }  
  
    mysql_free_result(result);  
  
    return balance;  
}
```

3.2. Security Measure:-

The security measures for a bank management system are crucial to safeguard sensitive financial information and maintain the trust of customers. Here are some key security measures commonly employed in bank management systems:

1. Encryption:

All data transmitted between users and the bank's servers should be encrypted using secure protocols such as TLS (Transport Layer Security).

Stored data, including customer information and transaction records, should be encrypted to protect it from unauthorized access.

2. Access Control:

Implement strong access controls to ensure that only authorized personnel can access sensitive information.

Use multi-factor authentication (MFA) to add an extra layer of security to user accounts.

3. Firewalls and Intrusion Detection/Prevention Systems:

Deploy firewalls to monitor and control incoming and outgoing network traffic.

Use intrusion detection and prevention systems to detect and respond to potential security threats.

4. Physical Security:

Ensure physical security measures are in place to protect servers, data centers, and other critical infrastructure.

5. Security Policies and Training:

Establish and enforce security policies and procedures for employees.

Provide regular training to staff on security best practices to prevent social engineering attacks and ensure awareness of potential risks.

6. Data Backups:

Regularly backup critical data and ensure that backup systems are secure.

Test data restoration processes to ensure that information can be recovered in case of data loss or a security incident.

Implementing a combination of these security measures can help create a robust and resilient bank management system that safeguards sensitive financial information and maintains the integrity of banking operations.

4.Additional Features:

A Bank Management System typically involves a range of features to efficiently handle various banking operations. Additional features can enhance the system's functionality, security, and user experience. Here are some additional features that could be considered for a Bank Management System.

1. **Biometric Authentication:** Implementing biometric authentication methods such as fingerprint or facial recognition can enhance security and streamline the login process for users.
2. **Multi-factor Authentication (MFA):** Adding an extra layer of security by incorporating multi-factor authentication can protect user accounts from unauthorized access.
3. **Mobile Banking App:** Developing a mobile banking application allows customers to access their accounts, perform transactions, and manage finances on the go.
4. **Personal Finance Management (PFM):** Including features that help users track and manage their expenses, set financial goals, and create budgets can enhance customer satisfaction.
5. **Alerts and Notifications:** Providing real-time notifications for transactions, account balances, and other important events helps users stay informed and monitor their financial activities
6. **Automated Customer Support:** Implementing chatbots or virtual assistants for customer support can enhance user experience by providing quick responses to common queries and issues.
7. **Card Management:** Allowing users to manage their debit/credit cards, including setting spending limits, activating/deactivating cards, and reporting lost or stolen cards, enhances user control and security.
8. **Online Loan Application and Management:** Facilitating the entire loan application process online, including document submission and status tracking, can streamline the lending process.

The specific features to include would depend on the bank's size, target audience, and the regulatory environment in which it operates. Additionally, it's crucial to prioritize security and compliance when implementing new features in a Bank Management System.

Explanation:-

1. Starting from the main() function, we are declaring a variable to know which specific operation the user wants to perform into the Student Record Management Systems , but on the user's command, it will keep getting executed always. After that, we ask the user to enter the operation they want to perform.



5.Conclusion:

- In this article, we have built a mini Bank Management System in C language using the concepts of arrays.
- In this project, the user can add a new student's record, view student details, search for a specific student's record, exit and more.
- We built it by creating individual functions for each possible operation, thus improving knowledge of the code.
- We have also tried to include information deletion and information editing.



6.References:

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Workload Matrix:

Name	NSBM ID	Contribution
Ediriweera C.A.		100%
Hearath H.M.T.G.		100%
Jayasinghe K.U.		100%
Weerasinghe E.M.L.C		100%
Weerasinghe M.G.H.		100%

THANK YOU!!!!.....